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EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS | Las Vegas

Building Angular PWAs: The Overlooked Path to Modern Mobile Apps

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Level: Introductory to Intermediate

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Agenda

- What is a PWA?
- Evaluating a PWA as an implementation
- Anatomy and Setup of a PWA
- Angular, Ionic and PWAs
- Service Workers
- Validating a PWA

Progressive Web Apps

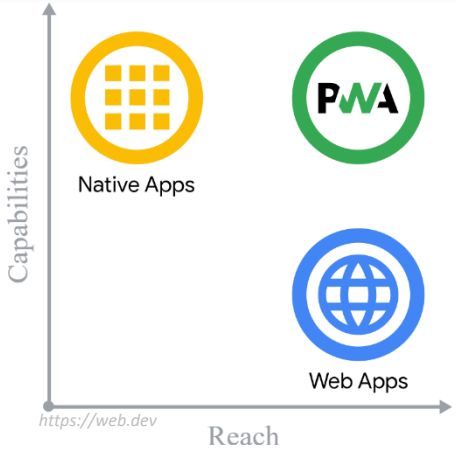
- Modern web stack
  - Built with web technologies
- Progressive enhancement
- Native like, mobile centric experience
- Installable
- Reliable / Offline capable

PWA Stack

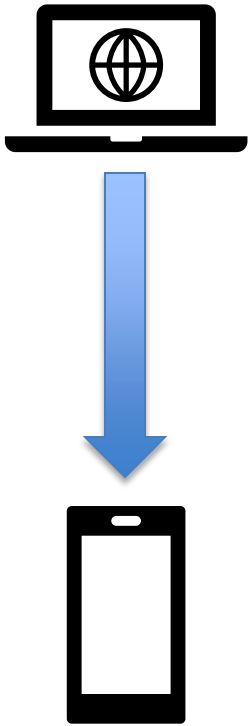
- HTML/CSS/JS/TS
- Web frameworks/libraries
  - Angular
  - React
  - Vue
- Browser native Web APIs
- Browser’s security model

Why PWA vs ____?

- PWA web development
  - Not a hybrid web framework
  - Not a native app using SDK
- Platform Agnostic
- Native-like experience with web implementation
- Require mobile or offline presence



How close to the metal?



Responsive Web

PWA

Hybrid Web

Hybrid Native

Native

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Characteristics of a PWA

- Fast and reliable
  - Loads quickly
  - Offline capable
- Installable
  - Secure / HTTPS
  - Installs service worker
- Optimized
  - Themes / colors
  - Icons
  - Splash screen
  - Mobile centric
  - HTTPS redirects

PWA Advantages

- No App Store Required
- Easily update like any other web app deployment
- Single codebase, multiple platforms
  - Multiple form factors
    - Responsive
- Leverage existing SPA
- Offline Capable

PWA Headwinds

"Your app should include features, content, and UI that elevate it beyond a repackaged website. If your app is not particularly useful, unique, or "app-like," it doesn't belong on the App Store. If your App doesn't provide some sort of lasting entertainment value or adequate utility, it may not be accepted."

– Apple App Store Review Guidelines
Minimum Functionality

PWA Disadvantages/Challenges

- Browsers
  - Safari and native capabilities
- iOS limitations
- Native access
  - Functionality not available on the web platform
- No store presence*
- Service workers can be more complex

PWA Mobile Alternatives

- Need more native power
  - Hybrid Web / Native
    - Capacitor (JS)
    - React Native (JS)
    - MAUI (C#)
    - Flutter (Dart)
  - Native
    - Swift (iOS)
    - Kotlin (Android)

Anatomy of a PWA

- Web App
- Web App Manifest
  - name, icons, start_url, display
- Service Worker
- Service Worker Configuration
- Assets / Icons
- Served via HTTPS
  - Azure
  - Firebase
  - http-server (npm)

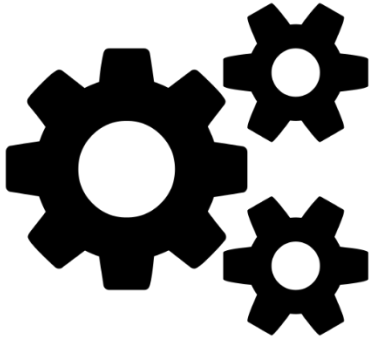
Angular and PWAs

- Angular includes a PWA library
  - @angular/pwa
- Use the CLI to add to an existing project
  - `ng add @angular/pwa`
  - Creates a service worker
  - Creates a manifest
  - Housekeeping to lace in new functionality

Ionic and PWAs

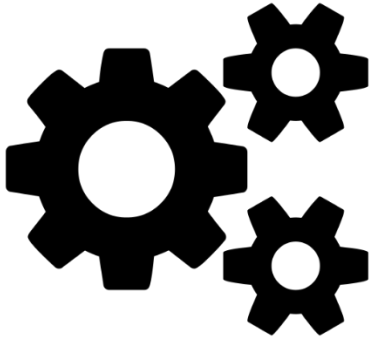
- Open-source framework for building cross platform apps
- Rich per-platform user experience
  - Single codebase
- Ionic CLI
  - Abstraction above Angular CLI
  - `npm install -g @ionic/cli`
  - `ionic start <enter>`
  - `or ionic init`
- Per Platform styling
  - https://ionicframework.com/docs/api/nav

Service Workers



- JS file running on separate thread
- Works as a proxy
  - Online = server
  - Offline = cache
- Service workers can be challenging
  - Angular Service Worker helps
  - Configurable via a .json file

Service Workers cont.



- Only work in prod builds
- Must use HTTPS
- Test with a local web server
  - Local web server or http web server
  - not `ng serve`
- Service Worker Lifecycle
- Advanced

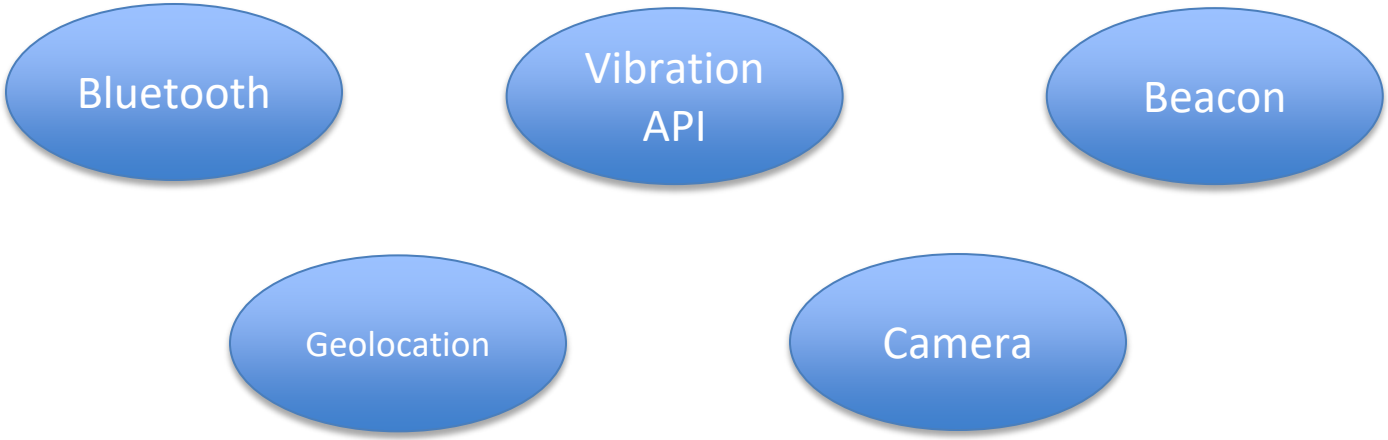
```
Current weather app version:  e642c087d092d7a03c0a12d1253f1b99dcb7ae31
Newest available app version:  61572397d24cdfcd57192187c4f2c2fd3970074b
Old weather app version prior to activation:  e642c087d092d7a03c0a12d1253f1b99dcb7ae31
New weather app version after activation:  61572397d24cdfcd57192187c4f2c2fd3970074b
```

NGSW Configuration

- Configuration file for service worker
  - Contains file and URL information
  - Cache and cache update policy
- Asset Groups
  - Resources that are part of the app
- Data Groups
  - Data resources / API requests
- Advanced Needs: Workbox

PWA and Hardware

- Web APIs
  - https://developer.mozilla.org/en-US/docs/Web/API



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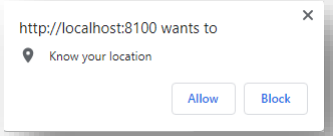
Geolocation API

- Provides location to web applications
- Requires user permission
- Leverages callbacks for success/failure
- `getCurrentPosition()`
- `watchPosition()`



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Validating a PWA

- Chrome Dev Tools
  - Application tab
    - Manifest
    - Service Worker
- PWA Builder
- ~~Lighthouse~~
  - Removed PWA check

Typical PWA Reports

Manifest

PWABuilder has analyzed your Web Manifest and your manifest is ready for packaging! We have identified recommended and optional fields that you can include to make your PWA better. Use our Manifest Editor to edit and update those fields.

Edit Your Manifest

Manifest Documentation →

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View Details

Required

- Manifest has icons field
- Manifest has name field
- Manifest has a short_name field
- Manifest has start_url field

Recommended

- Manifest has an app ID
- Manifest has launch_handler field
- Manifest has screenshots field
- Manifest has hex encoded background_color
- Manifest has description field
- Manifest has display field
- Icons have at least one icon with purpose any
- Manifest has orientation field
- Manifest has hex encoded theme_color

Optional

- Manifest has categories field
- Manifest specifies a default direction of text
- Manifest has iarc_rating_id field
- Manifest properly sets prefer_related_applications field
- Manifest has related_applications field
- Manifest has scope_extensions field
- Manifest specifies a language
- Manifest has scope field

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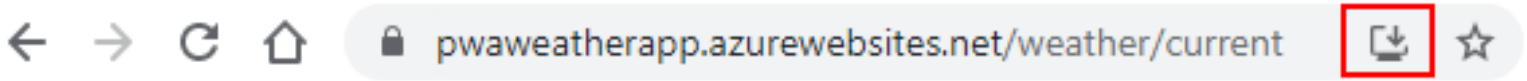
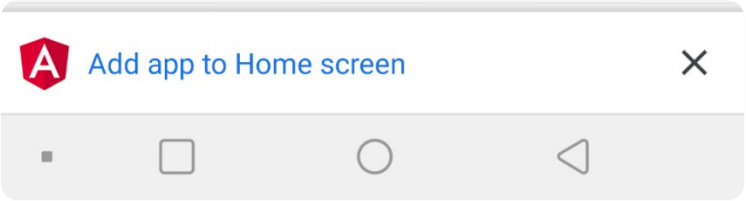
PWA Distribution Options



- Install from the browser
- Create an App Launcher
  - ‘+‘ icon
- Shows like an app on the OS
- Enterprise
- App Store
- PWA builder



Useful References

- GitHub Repo of this Code
  - https://github.com/AllenConway/PWAWeatherApp
- Progressive Web Apps
  - https://web.dev/explore/progressive-web-apps
  - https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps
- Angular Service Workers
  - https://angular.dev/ecosystem/service-workers/getting-started
- Ionic PWA
  - https://ionic.io/pwa
- Ionic CLI
  - https://ionicframework.com/docs/cli
- Web APIs
  - https://developer.mozilla.org/en-US/docs/Web/API
- Angular
  - https://angular.dev/
- TypeScript
  - https://www.typescriptlang.org
- PWA Builder
  - https://www.pwabuilder.com/

Session Survey

- Your feedback is very important to us
- Please take a moment to complete the session survey found in the mobile app
- Use the QR code or search for “Converge360 Events” in your app store
- Find this session on the Agenda tab
- Click “Session Evaluation”
- Thank you!

Thank you! Q&A

http://c1.staticflickr.com/1/28/65098350_b7bd96f38_b.jpg

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